

Creativity and Emergence of Specific Dance Movements Using Instructional Constraints

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Contact improvisation is a form of dance improvisation that involves 2 bodies in contact. Theoretical approaches most used to explain creative behavior in dance are insufficient to explain how dancers create new configurations of movements during improvisation. The aim of this study was to observe the effects that the manipulation of instructional task constraints has on the creative behavior of dancers while improvising, and also to analyze the spontaneous emergence of specific movements in relation to the task constraints imposed. Three dancers were video-recorded while dancing contact improvisation in duets lasting 480 s and with 3 different instructions (without instructional constraints, with pelvises as close as possible, or with pelvises as far apart as possible). Dance data were recorded using an observational instrument and subsequently analyzed by means of a principal components analysis and the dynamic overlap order parameter (measure capable of detecting the rate and breadth of exploratory behavior on different time scales). The coupling strength between duets was examined by a cross-correlation analysis. The effects of instructional as well as nonspecific constraints were then analyzed by a 1-way analysis of variance. Results revealed that instructional constraints had a significant effect on the type of configurations performed by the dancers, as well as on their creative behavior. Of the 3 conditions, the second instructional constraint (IC2) induced a significantly greater variety of action patterns.

Keywords: instructional constraints, motor creativity, emergence, contact improvisation

Methods for enhancing creative behavior in sport or physical activities have received much attention in recent years because of the associated performance benefits. Traditionally, performance optimization methods have been implemented to identify the set of movement parameters that might maximize outcomes for specific elite athletes (e.g., Bartlett, 2007). However, it is also necessary to find ways of improving performance outcomes in team sports or physical practices that require the constant use of creativity. Despite this, there remains a need to establish a thorough formal model for studying and explaining creative behavior in sport or dance (Hristovski et al., 2013).

Dance is one area in which the question of creativity has recently attracted considerable research interest, because creativity is considered to be a crucial aspect of all artistic activity (Bachner-

Melman et al., 2005; Banaji, Burn, & Buckingham, 2006; Chappell, 2007; Connell, 2009; Craft, 2005; Tokinan & Bilen, 2011; Torrents, Castañer, Dinušová, & Anguera, 2013). However, whether in sport or artistic activity, the creative process is usually related with cognition and with a linear process that is driven by “possibility thinking,” a process that begins with imaginatively asking “what if?”, and then moves on to performing the action. Although this approach can be useful for many situations related with dance composition, it is insufficient to explain how dancers create new configurations of movements during improvisation. A further point is that to define what is creative or not, researchers propose two essential characteristics of a creative product or behavior (Amabile, 1995; Kasof, 1995; Mumford & Simonton, 1997; Runco, 2004; Sternberg & Lubart, 1996): novelty or originality, and value (utilitarian and/or esthetic). In dance, however, we are not only interested in the product but also in the process or the exploratory behavior (Drazin, Glynn, & Kazanjian, 1999; Sharp, 2004). This behavior is the central characteristic of improvisation, a tool that is now widely used by dance teachers and choreographers (Biasutti, 2013; Lavender & Predock-Linell, 2011) to encourage their students or dancers to explore what might emerge from a set of given instructions. The most interesting outcomes of this process can be used for composition (Buckwalter, 2010). It is interesting that there are specific dance styles based on improvisation; for example, social ballroom dance, tap dance, contemporary improvised dance, free dance, and contact improvisation (CI).

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CI can be defined as a movement form that is improvisational in nature and involves two bodies in contact (Sidall, 1997). The improvisational characteristics of CI are such that the generation of movements is not based on fixed and standardized movements or techniques, because this dance form requires a body that responds to the physical exchange of weight and contact (Albright, 2003; Novack, 1990). What distinguishes CI from other modern dance forms is the sharing of a constantly moving point of contact, as well as the fact that movements arise from a relationship with a partner rather than from a preconceived ideal (Pallant, 2006). Therefore, the movements that appear during CI are not foreseeable and are constantly changing.

The teaching/learning process of CI, as well as of creative or improvisation dance classes, is based on exploration using guided-discovery styles. Teachers propose situations of practice with the aim of improving the sensitivity of performers to the stimuli produced by the bodies of dancers dancing together. Expertise in dancing CI is achieved when performers can adapt their dance to any new partner with different abilities and styles, maintaining the fluidity and variety of the dance, and remaining constantly aware of the present. In this sense, Buckwalter (2010) proposes that “Contact is unlike most dance forms in that the more the form becomes fixed in its vocabulary and techniques of execution, the less successful it is in fulfilling its initial task of putting the body off its guard. As dancers become familiar with the ins and outs of the basic movements that arise, they may create a flowing dance of negotiating how they will shift from one maneuver to another” (p. 42). This process involves the practice of as many situations of dance as possible, and the teacher’s role will be to create the right contexts to induce the emergence of new movements that have not been practiced before (Kaltenbrunner, 2004). In activities of this kind, one always faces the dilemma of whether the creative process is best supported by providing freedom or imposing constraints (Craft, 2005). Chappell (2007) considers that the answer depends on whether the creative source is prioritized within the students *inside out*, or within dance knowledge or the teacher working *outside in*. Our proposal here, however, is focused on using constraints to enhance the flexibility of outcomes or for enabling the emergence of specific motor responses. As other authors propose in art, to produce a creative solution, constraints preclude existing responses, but also promote novel, often opposite ones, and require that a response differ from some number of prior responses increments variability (Stokes, 2001, 2008). In this sense, CI may be viewed as a learning-through-exploration method. Constraints will be used with a learner-centered approach, using open-ended problem-solving methods.

Hristovski, Davids, and Araújo (2009), and Hristovski, Davids, Araújo, and Button (2006), in an empirical study based on the principles of coordination dynamics and using the framework of complex systems theories (see Kelso, 1995), showed how the manipulation of task and personal constraints may enable the emergence of innovative and functional tactical behaviors in athletes during sport performance. Complex systems approach helps us to describe, explain, and predict how patterns of coordination form, emerge, evolve, and change in terms of dynamical laws or rules (Kelso, 2009). For example, under a specific mix of ecological constraints, performers in boxing generated certain behavioral states. It was observed that changing the set of task constraints led to spontaneous and novel behaviors; in this case, specific striking

techniques that had not been practiced or taught before. These observations of the emergence of human tactical innovative behaviors are important from the perspective of Boden’s (1996) definition of “transformational creativity,” because they explain how, when manipulating constraints, a transformation within a neurobiological action system may be enabled without reference to abstract rule-governed behaviors. It should be noted that this approach has been applied to skill acquisition in sport (Araújo, Davids, Bennett, Button, & Chapman, 2004; Chow et al., 2006; Davids, Araújo, Shuttleworth, & Button, 2003; Davids, Renshaw, & Glazier, 2005; Newell, 1986), where it has generated considerable interest because of its clear applications, especially in relation to team sports. The dynamic and nonlinear relationship between players, teams, and the game context gives rise to patterns and transitions among them (Araújo, Davids, & Hristovski, 2006; Passos et al., 2008). These transitions occur as a result of spontaneous self-organization, and this has important consequences for the training and teaching/learning processes. Traditionally, coaches constructed partial and separate training exercises for technical, physiological, and tactical aspects, and in some cases there was no correspondence with actual game situations (Mitchell & Oslin, 2007). However, coaches need to develop training situations that are representative of actual competition to achieve a better coupling between perception and action in the game (Araújo, Davids & Passos, 2007). A further point is that individual sports training methodology usually involves a fragmentation of body systems and the reproduction of skills based on an ideal and external model. Something similar occurs in the dance context, at least in the processes of teaching specific dance skills. Traditionally, learning specific dance motor skills has involved the copying of models, reinforced by metaphorical images and verbal descriptions (Harbonnier-Topin & Barbier, 2012; Spohn & Spickard, 2012). The question, therefore, is whether dance teachers could use other strategies to teach motor skills, without having to copy models through the manipulation of task constraints. Might a manipulation of this kind influence the creative behavior of dancers?

The aim of this study is to examine the effects that the manipulation of instructional task constraints has on the emergence of exploratory behavior in dancers while improvising, and also to analyze the spontaneous emergence of specific movements in relation to the task constraints imposed.

Method

Three dancers of contemporary dance (two women and one man: 28 ± 4.6 years, 60.33 ± 8.4 kg), all with more than 5 years of experience in CI, participated voluntarily in the study. They had danced together before, but just occasionally and for recreational purposes. Participants warmed up individually. Each participant had to dance with the other two dancers in duets lasting 480 s, and in a limited space of 12×12 m. In the first duet danced by each pair there were no special instructional constraints (without instructional constraints, WIC), except those produced by contact with the partner (i.e., visual and haptic information, and the force of gravity). Afterward, the dancers danced two more times with each partner under two different instructional constraints (ICs):

- IC1: When dancing, try to keep your pelvis as close as possible to your partner.

- IC2: When dancing, try to keep your pelvis as far away as possible from your partner.

Data Acquisition and Preparation

The nine duets were video-recorded, and video images were analyzed by two independent observers, both experts in dance. For all duets, the observers took into account the movements performed by each of the two dancers, such that a total of 18 duet sessions were analyzed. Sequences of actions/postures were analyzed to determine their structural and dynamical characteristics. Action or posture skills and movements were defined on a coarse-grained scale. A total of 49 categories, corresponding to elementary movement/posture skill components (Table 1), were taken into

account; these categories covered support/contact characteristics and directions and planes of movement of body segments, in accordance with previous research using this kind of observational methodology (Castañer, Torrents, Anguera, Dinušová & Johnson, 2009; Torrents, Castañer, Dinušová & Anguera, 2010). A value of 1 was ascribed to the active category and a value of 0 to the inactive one. This enabled the formation of a temporal 49×480 binary matrix with a time resolution of 1 s. Each 1-s window was defined as a 49-component binary vector (column) representing the full action configuration during the same time interval. It is important to note that each full action configuration ($L_{\max} = 49$) was composed of combinations of skill and movement of shorter length ($1 < L < L_{\max}$), which in turn

Table 1

Observational Instrument Defined on a Coarse-Grained Scale Based on Action or Posture Skills and Movements

1–6: <i>Support stability skills involving the floor:</i> Actions of stability or balance using the floor that can be supported by . . .	Upper limbs (one or two)	SSU
	Lower limbs (one or two)	SFL
	Head	SFH
	Pelvis	SFP
	Torso	SFT
	Back	SSB
7–12: <i>Support stability over a partner:</i> Actions of stability or balance supported over the . . . of the partner	Upper limbs (one or two)	SPU
	Lower limbs (one or two)	SPL
	Head	SPH
	Pelvis	SPP
	Torso	SPT
	Back	SPB
13–16: <i>Axial stability skills:</i> Turns around the . . .	Longitudinal axis of the body	TLA
	Horizontal transverse axis of the body	TTA
	Horizontal anteroposterior axis of the body	TAA
	A combination of axes	TCA
17: Jumping		J
18: Being lifted or sustained by the partner		BE
19–20: Level changing from . . .	Middle to down or falling	CMD
	Down to middle or rising from the floor	CDM
21–24: Locomotor skills involving . . .	Walking	WLK
	Movement on all fours	MAF
	Rolling	RL
	Sliding	SL
25: Receiving the partner		RP
26: Colliding with the partner		CLP
27: Leading the partner		CNP
28: Lifting or sustaining the partner		ESP
29: Avoiding or escaping from the partner		EP
30–47: Positions or actions of the different parts of the body:	One leg is bent	OLB
	Both legs are bent	BLB
	One leg is moving	OLM
	Both legs are moving	BLM
	One arm is bent	OAB
	Both arms are bent	BAB
	One arm is moving	OAM
	Both arms are moving	BAM
	One arm is relaxed, following gravity line	OAR
	Both arms are relaxed	BAR
	Body is almost aligned	BA
	Body is bent forwards	BBF
	Body is bent backwards	BBB
	Body is bent to the right	BBR
	Body is bent to the left	BBL
	Body is moving	BM
	Body is inverted (more than 45°)	BI
	Head is moving	HM
48: There is a change in the direction of the movement		CDM
49: There is a change in the global position of the body		CBP

comprised elementary skill/movement categories, each with a length $L_{\min} = 1$.

Data Analysis

Analysis of the movement improvisation structure. A principal component analysis (PCA) under Direct Oblimin rotation with $\delta = 0$ (Westerhuis, Kourti, & Macgregor, 1998) was performed on the 49 (category) $\times$ 480 (time point) binary matrices defining movement/posture configurations (for the suitability of using PCA on binary variables, see Jolliffe, 2002, p. 339). This was done with the aim of detecting the structure of the dancer's action landscape and to consider the possibility of a higher order structure in the data, as predicted by the Hristovski et al., 2011, 2013, model. Because oblique rotations are more general and contain the orthogonal rotations as a specific case (e.g., Fabrigar, Wegener, MacCallum, & Strahan, 1999), using the Direct Oblimin rotation was an obvious objective way to test the model. The number of significant principal components (PCs) was determined by identifying those that accounted for $\geq 80\%$ of the explained variance. A one-way analysis of variance (ANOVA) was performed to detect possible effects of ICs on the total number of times that the dancers performed each action/posture. The significance level was set at .05.

Analysis of the exploratory dynamics of duets. The average dynamic overlap, $\langle q_d(t) \rangle$, was calculated as an average cosine auto-similarity of the overlaps between $L_{\max}$ configurations with increasing time lag to determine the dynamic properties of a dancer's complex movement patterns (for more details, see Hristovski et al., 2013). This measure captures the average similarity of configurations at ever-increasing time distances (i.e., time lags) from each other. Hence, it is capable of detecting the rate and breadth of exploratory behavior on different time scales. The average dynamic overlap was fitted by the following equation, which is derived for systems with an intricate hierarchical structure (see Sibani & Dall, 2003, among others):

$$\langle q_d(t) \rangle = (1 - q_{\text{stat}})t^{-\alpha} + q_{\text{stat}} \quad (1)$$

where q_{stat} is the asymptotic (i.e., stationary) value of the dynamic overlap, t is the time lag, and α is the dynamic exponent. q_{stat} detects the long-term exploratory breadth of the dancer, and α the rate of exploration. In Figure 1, it can be seen that in the long-term (time scales of tens and hundreds of seconds) the average dynamic overlap, $\langle q_d(t) \rangle$, converges to a stationary value, $\langle q_{\text{stat}} \rangle$. However, it initially shows a descending trajectory with a slope defined

by the exponent α , corresponding to the short-term (time scale of seconds) exploratory breadth.

Hence, the average dynamic overlap, $\langle q_d(t) \rangle$, captures both the short- and long-term exploratory behavior of dancers through the values of parameters α and q_{stat} , respectively. The fit between the theoretical prediction and the experimental data was assessed by the quasi-Newton method of least squares estimation, with the goodness-of-fit being based on the percentage of explained empirical variance.

A one-way ANOVA was then conducted to compare the effect of the three different ICs on the mean long-term exploratory breadth, $\langle q_{\text{stat}} \rangle$, and the short-term exploratory breadth, α . The alpha level was set at $p < .05$, and Cohen's d coefficients were calculated to establish the effect sizes.

Analysis of duet coupling dynamics. To examine the coupling dynamics of the duet, we conducted a cross-correlation analysis of the Hamming distance time series. Hamming distance time series provide information about the size of individual reconstructions in each time point, and hence, cross-correlating the time series tells us about the time profile and the strength of duet couplings.

Results

Analysis of interobserver agreement yielded a kappa coefficient of 0.93. This value provides a satisfactory guarantee of data quality.

There was a significant effect (at the $p < .05$ level) of the instruction on the frequency of performance of some movement/postures for the three constraints. In terms of the movements performed, the first instructional constraint (IC1) produced significantly more movements involving (a) contact with the partner via the pelvis, torso, and back, $F(2, 15) = 59.16$, $p < .01$; 3.76, $p < .05$; 13.04, $p < .01$; (b) more lifts (being lifted or sustained): $F(2, 15) = 10.67$, $p < .01$; and (c) lifting or sustaining the partner, $F(2, 15) = 8.12$, $p < .01$. The second instructional constraint (IC2) produced significantly more skills involving (a) contact with the partner via the head, $F(2, 15) = 12.32$, $p < .01$; (b) more jumps, $F(2, 15) = 5.74$, $p < .01$; and (c) more walking, $F(2, 15) = 7.20$, $p < .01$ (Table 2).

The Primary-Level Movement Structure

Under the Kaiser-Guttman criterion, the PCA initially revealed between 22 and 28 primary PCs in each file. As noted by several authors (e.g., Fabrigar et al., 1999; Velicer & Jackson, 1990), the Kaiser-Guttman criterion often leads to overfactoring; hence, only the set of primary PCs accounting for more than 80% of the total variance were taken into account (Jolliffe, 2002), resulting in between 8 and 11 PCs for each file to analyze. The higher-order PC analysis results revealed very interesting structure; however, because of spatial reasons, we restrict ourselves to comments of the first-order component scores.

In general, the results from the component scores analysis show that certain movements attract the system and appear with a high frequency, while others are statistically rare and short-lived reconstructions that constitute fluctuations. Movements appearing with high frequency loaded predominantly on the first PC, capturing most of the variance, although they also had high component

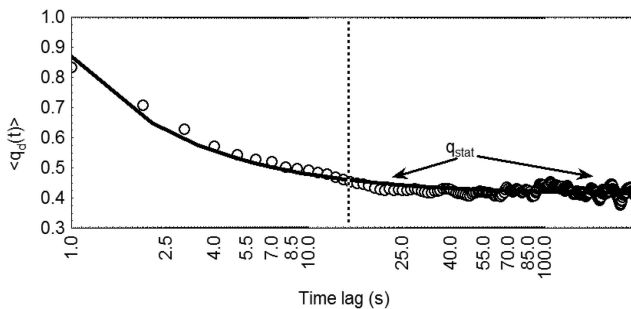


Figure 1. An example of quasi-Newton least squares fit of the theoretical curve (see Equation 1) and experimental data (see explanation in the text).

Table 2

Results of the One-Way Analysis of Variance (ANOVA) Conducted to Compare the Effect of the Three Different Instructional Constraints on Frequency of Emergence of Movement Categories

Variable	WIC	IC1	IC2	F	p
Support stability skill with the partner and the head	17.00	4.17	25.00	12.32	<.01
Support stability skill with the partner and the pelvis	82.67	141.83	11.17	59.16	<.01
Support stability skill with the partner and the torso	45.33	53.67	32.83	3.76	<.05
Support stability skill with the partner and the back	71.83	89.67	44.67	13.04	<.01
Being lifted or sustained by the partner	27.17	44.67	20.50	10.67	<.01
Jump	1.33	0.83	3.00	5.74	<.01
Walking	23.50	5.50	31.50	7.20	<.01
Lifting or sustaining the partner	23.83	37.00	17.17	8.12	<.01
Both legs are bent	41.67	67.33	48.50	11.83	<.01
Both arms are bent	37.50	58.83	57.50	3.73	<.05
Body is bent backwards	9.33	18.67	12.00	5.17	<.05
Position of the body is inverted by more than 45°	2.33	3.17	9.00	6.31	<.01
There is a change in the direction of the movement	29.83	5.50	21.00	8.69	<.01

Note. WIC = without instructional constraints; IC1 = first instructional constraint; IC2 = second instructional constraint. Alpha level was set at $p < .05$.

scores on several other PCs that play the role of a “bridge” and are responsible for the correlations between first-level PCs. For example, leading or lifting the partner using the foot-floor surface as support, support of the partner with the lower and upper limbs, and use of the trunk/torso/back/pelvis surfaces as support were present, with only quick interruptions, over the whole observation time scale.

Table 3 shows the first four PCs of a dancer in the three conditions (WIC, IC1, and IC2) with the same partner. The categories that have a high positive value usually appear together. The categories that have a high negative value usually appear in other PCs, but never or almost never with the categories of the PC analyzed. In the example shown, the movements corresponding to the first PC involve leading the partner using the foot-floor surface as support, and support of the partner with the upper limbs, head and back when dancing WIC, with the pelvis and back when dancing under IC1, and with the upper limbs when dancing under IC2. Arm movement and body alignment are also present in the three conditions. With the other PCs, clearer differences emerge, because the categories “being lifted/sustained by the partner” or “lifting/sustaining the partner” are only present in the first PC when dancing WIC or under IC1, whereas this is not the case with IC2. By contrast, turns or walking are only present when dancing under IC2.

The analysis of component scores of first-level PCs for all trials showed that when participants danced WIC they used the foot/hand-floor surface as support, and support with the partner involving various body parts. Leading the partner was clearly the most commonly used movement, followed by being sustained/lifted or lifting/sustaining. Walking was also frequent. Lifting/sustaining the partner was performed using the support of pelvis and back with the partner and foot-floor surface support, while being lifted was performed with pelvis and torso or pelvis and back over the partner.

When participants danced under IC1 they used the foot/hand-floor surface support and support with the partner involving the pelvis and back. Leading the partner was the most commonly used movement, followed by being sustained/lifted or lifting/sustaining. In this case, lifting/sustaining the partner appears in the first PC of two trials, whereas walking or turning do not appear in the first

PCs. Lifting/sustaining the partner was performed using the support of pelvis and back, while being lifted was performed with pelvis and torso over the partner.

When participants danced under IC2 they used the foot-floor surface support, whereas hand-floor surface support was not widely used. The surfaces used for contacting the partner were more varied here, although the most commonly used were the upper limbs. Leading the partner was again the most widely used movement, it being present in the first five PCs (leading the partner was performed 494 times when dancing WIC, 342 with IC1, and 335 under IC2; these differences were not significant). In one of the trials, lifting/sustaining the partner was present in the first PC, but in the other trials this movement did not appear until the third PC or later. Walking was also present in some of the first PCs, as was turning. Most lifts were performed when the upper limbs and other body parts were in contact with the floor, whereas the lower limbs were touching the floor when being lifted/sustained. This means that when dancing under IC2, dancers sustained their partner in different positions, especially in floor positions such as sitting, landing, or on all fours, but they did not lift them at all.

Exploratory Dynamics

Figure 2 shows an example of the evolution of cosine auto-similarity; that is, the average dynamic overlap ($\langle q_d(t) \rangle$), of one dancer dancing under the three ICs. Equation 1 showed a remarkably good fit to all time series of the dynamic overlap, with average explained variance $R^2 = 0.91 \pm 0.01$. If the dancers had remained in a single posture throughout the observation time of 480 s, then the average dynamic overlap would be a constant equal to 1 (i.e., $\langle q_d(t) \rangle = 1$) for all time lags. On the other hand, if, during the performance, the dancers explored distant configurations of the whole state space, then the average dynamic overlap ($\langle q_d(t) \rangle$) would drop exponentially and quickly to zero for the observation time of 480 s. On average, movement configurations separated in time by more than 15–20 s converged to a stationary average overlap of $\langle q_{stat} \rangle \gg 0$ (see Equation 1). Because the dynamic overlap forms a plateau at values far from zero, the system does not explore arbitrarily distant movement configurations but, rather, dwells in a relatively large, albeit limited, region

Table 3

First Four Principal Components of a Dancer in the Three Conditions (WIC, IC1, and IC2) With the Same Partner

	WIC PC1	IC1 PC1	IC2 PC1	WIC PC2	IC1 PC2	IC2 PC2	WIC PC3	IC1 PC3	IC2 PC3	WIC PC4	IC1 PC4	IC2 PC4
SSU	32.98	39.00	38.51	14.24	9.06	9.82	9.03	7.97	7.29	5.19	6.39	5.59
SFL	-0.63	-1.09	-0.40	0.56	-1.87	0.89	0.91	1.45	0.21	0.15	-1.67	-0.36
SFH	1.96	2.34	1.62	-2.05	-2.05	1.25	2.36	0.89	1.56	0.73	-1.57	1.96
SFP	-0.54	-0.37	-0.65	0.60	0.34	0.09	-0.31	-0.39	-0.70	-0.36	0.59	-0.24
SFT	-0.44	-0.57	-0.90	0.59	0.03	1.12	-0.30	-0.58	0.55	-0.57	-0.70	-0.67
SSB	-0.61	-0.64	-0.18	0.55	0.28	-0.34	-0.30	-0.46	-0.39	-0.59	0.29	-0.30
SPU	-0.46	-0.43	-0.65	0.66	0.38	0.20	-0.57	-0.48	-0.52	-0.39	0.59	-0.27
SPL	2.47	-0.76	2.76	-1.29	0.80	0.74	-0.25	-0.08	0.33	-0.48	1.36	0.17
SPH	-0.32	-0.54	-0.49	0.58	0.56	-0.36	-0.17	2.66	-0.29	-0.52	-0.49	-0.44
SPP	1.13	-0.20	-0.18	0.38	0.31	-1.08	-0.61	-0.47	1.19	1.68	0.44	-1.05
SPT	-1.33	2.16	-0.36	0.44	-1.82	-0.40	1.43	2.30	-0.12	2.47	-0.90	-0.40
SPB	0.06	-0.11	-0.23	-0.43	1.14	-0.89	-0.70	2.81	-0.15	-0.47	0.85	-0.37
TLA	2.13	2.63	0.17	-0.19	-2.33	-0.69	1.93	-0.90	-0.14	3.49	-0.57	-0.03
TTA	-0.36	0.42	-0.25	-2.10	0.62	1.02	-0.13	-0.54	-0.04	-0.46	0.85	1.88
TAA	-0.26	-0.40	-0.13	0.54	0.35	-0.48	-0.50	-0.45	-0.21	-0.47	0.45	-0.36
TCA	-0.26	-0.41	-0.18	0.54	0.34	-0.25	-0.50	-0.48	-0.16	-0.47	0.26	-0.34
J	-0.31	-0.47	-0.13	0.51	0.41	-0.48	-0.53	-0.49	-0.21	-0.51	0.35	-0.36
BE	-0.26	-0.41	-0.12	0.54	0.36	-0.44	-0.50	-0.43	-0.24	-0.47	0.44	-0.28
CMD	-0.34	-0.55	-0.14	0.71	0.76	-0.39	-0.79	2.31	-0.24	2.74	0.80	-0.35
CDM	-0.33	-0.44	-0.19	0.59	0.31	-0.27	-0.31	-0.53	-0.43	-0.43	0.22	-0.23
WLK	-0.33	-0.19	0.08	0.58	0.46	-0.30	-0.70	-0.16	-0.10	-0.23	0.59	-0.33
MAF	-0.48	0.20	0.19	-2.19	0.52	-0.16	-0.55	-0.43	-0.61	-0.49	0.70	2.46
RL	-0.26	-0.40	-0.13	0.54	0.35	-0.48	-0.50	-0.45	-0.21	-0.47	0.45	-0.36
SL	-0.44	-0.32	-0.13	0.55	0.37	-0.48	-0.43	-0.54	-0.21	-0.58	0.58	-0.36
RP	-0.24	-0.40	-0.13	0.57	0.35	-0.48	-0.50	-0.45	-0.21	-0.40	0.45	-0.36
CLP	-0.26	-0.40	-0.11	0.54	0.35	-0.51	-0.50	-0.45	-0.22	-0.47	0.45	-0.36
CNP	-0.26	-0.40	-0.13	0.54	0.35	-0.48	-0.50	-0.45	-0.21	-0.47	0.45	-0.36
ESP	2.61	3.05	2.92	-1.83	0.53	1.59	-1.29	-0.19	0.55	-0.91	-1.74	-0.78
EP	-0.75	-1.08	-0.62	0.49	-2.39	-1.20	2.77	-0.82	0.13	-0.93	0.82	0.07
OLB	-0.26	-0.40	-0.13	0.54	0.35	-0.48	-0.50	-0.45	-0.21	-0.47	0.45	-0.36
BLB	-0.29	-0.28	-0.09	0.51	0.42	-0.61	-0.51	0.29	-0.13	-0.52	0.13	-0.57
OLM	0.75	-0.77	-1.08	0.75	-2.61	1.34	3.08	-0.46	2.98	0.67	-1.23	-0.62
BLM	-0.26	-0.23	-0.06	0.54	0.20	-0.53	-0.50	-0.17	-0.18	-0.47	0.06	-0.44
OAB	-0.58	1.26	1.26	-2.83	0.93	1.31	0.00	0.42	-1.70	2.00	-0.75	3.69
BAB	-0.69	-0.52	-0.88	-0.65	0.18	-0.32	-0.30	-0.61	-0.70	-0.44	0.68	-0.40
OAM	-0.07	0.29	1.72	0.10	0.13	-1.56	1.54	1.51	4.13	-0.44	0.16	2.10
BAM	-0.69	-0.70	-0.88	-0.30	0.07	-0.29	-0.02	-0.72	-0.53	-0.43	0.70	-0.27
OAR	3.18	1.48	3.50	0.31	0.24	2.59	-0.04	0.12	-0.47	-0.77	-1.67	-0.64
BAR	-0.70	-0.35	-0.74	-0.33	0.37	-0.48	-0.72	-0.47	-0.53	-0.51	0.58	-0.35
BA	-0.72	0.56	-1.11	0.25	-0.20	-0.50	-0.57	-0.22	-0.57	1.51	0.52	0.23
BBF	2.18	2.65	1.25	-1.91	-2.03	-1.64	0.38	-0.78	-1.14	0.18	1.39	2.63
BBB	-0.70	-0.32	-0.63	0.62	-2.20	-0.25	2.46	2.75	2.23	-0.56	0.39	-0.45
BBR	-0.31	-0.56	-0.01	0.48	0.63	-0.25	-0.80	-0.85	-0.64	1.51	-0.02	-0.78
BBL	-0.26	-0.38	-0.22	0.54	0.39	-0.39	-0.50	-0.49	-0.10	-0.47	0.30	-0.41
BM	-0.27	-0.40	-0.13	0.56	0.43	-0.48	-0.50	-0.42	-0.21	-0.40	0.35	-0.36
BI	-0.59	-0.74	-0.54	0.77	0.64	2.22	-0.09	-0.41	-0.57	-0.13	-3.10	-0.83
HM	-0.20	-0.35	-0.13	0.54	0.26	-0.50	-0.36	-0.35	-0.22	-0.47	0.48	-0.38
CDM	0.57	0.46	-1.43	-0.35	1.11	2.62	0.29	-0.15	1.37	0.65	-1.13	0.84
CBP	-0.25	-0.29	-0.50	-2.49	0.36	-0.53	-0.17	-0.45	-0.53	-0.33	0.46	0.47
	-0.77	-0.61	-0.52	0.79	0.53	1.98	-0.17	-0.31	-1.21	-0.22	-3.06	-0.66

Note. WIC = without instructional constraints; IC1 = first instructional constraint; IC2 = second instructional constraint; PC = principal components. See Table 1 for remaining abbreviation definitions. Positive values >1 are in bold, while negative values <1 are in italics.

of performer–environment state space. Nevertheless, the plateau values ($\langle q_{stat} \rangle$) much smaller than 1 indicate that the system shows relatively high exploratory (i.e., creative) fluency.

Figure 3 presents the means across all trials of the stationary overlap order parameter, $\langle q_{stat} \rangle$. There was a significant effect of ICs on the long-term exploratory breadth of duets, $F(2, 15) = 11.10$; $p < .001$. The average values obtained for the three conditions were as follows: when dancing WIC, $\langle q_{stat} \rangle = 0.459 \pm 0.035$; when dancing under IC1, $\langle q_{stat} \rangle = 0.491 \pm 0.021$; and

when dancing with IC2, $\langle q_{stat} \rangle = 0.416 \pm 0.026$. The long-term exploratory breadth was significantly higher (lower $\langle q_{stat} \rangle$) when dancing with IC2 compared with dancing under both WIC ($p < .04$; Cohen's $d = 1.39$) and IC1 ($p < .0009$; Cohen's $d = 3.17$). The long-term exploratory breadth was lower (higher $\langle q_{stat} \rangle$) when dancing with IC1, but it was not significantly different from the $\langle q_{stat} \rangle$ of dancing without ICs ($p < .15$), although the corresponding effect size was nevertheless high (Cohen's $d = 1.11$). Generally speaking, the mean value of $\langle q_{stat} \rangle$ when danc-

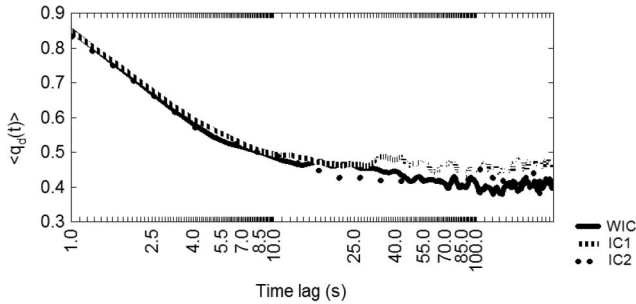


Figure 2. An example of evolution in the average dynamic overlap for three different instructional constraints of a duet. WIC = without instructional constraints; IC1 = first instructional constraint; IC2 = second instructional constraint.

ing WIC was located between the extremes of IC1 and IC2, meaning that for most combinations of pairs of partners it had a kind of in-between exploratory breadth.

The ICs—WIC, IC1, and IC2—had no significant effect on the short-term exploratory breadth, $F(2, 15) = .15$; $p < .86$.

Duet Coupling Dynamics

The cross-correlation of Hamming distances that measured the size of reconfigurations showed consistently weak but significant lag 0 (synchronous reconfigurations) and lag 1 (leading-following reconfigurations) correlations. Average lag 0 correlations had a value of $r = .35 \pm 0.08$, while lag 1 cross-correlations gave $r = .30 \pm 0.08$. A typical example is shown in Figure 4. It is interesting to note that lag 1 cross-correlations were exclusively the property of one duet under different ICs. These results show that two duets were preferentially more adhered to synchronous reconfigurations, while one duet was preferentially adhered to short-term delayed reconfigurations.

Discussion

The results corroborate the predictions of a topological and temporal structure of relatedness between configurations emerging during CI (Hristovski et al., 2013). The dimension reduction showed that instructional task constraints induce a correlated structure within the movement state space of dancers, with IC1 creating more correlated and, therefore, less explorative movement structures.

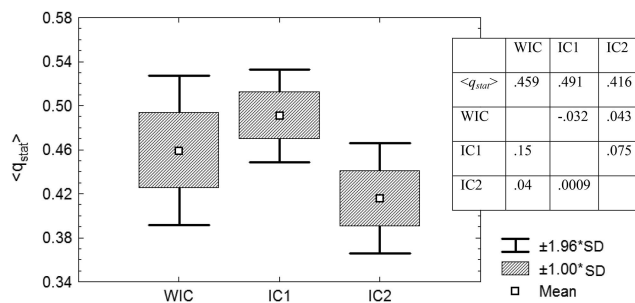


Figure 3. Mean of all trials of the overlap order parameter $\langle q_{stat} \rangle$.

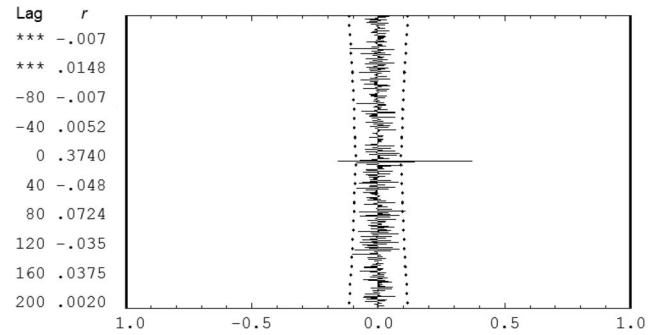


Figure 4. A typical profile of a cross-correlation function of contact improvisation (CI) between reconfigurations of duet dancers.

ICs had a significant effect on the type of configurations performed by the dancers. When dancers were asked to dance while keeping their pelvis as close as possible to their partner, they tended to lift their partners or let themselves be lifted by them, this likely because lifts are easier to perform when the partner's center of mass is close to the body of the lifter (Fortin, Harrington, & Langenbeck, 1997). IC1 constrained the system, changing the exploratory breadth of the dancers. Different configurations related with lifts will thus be more likely to occur here than with IC2 or WIC.

Although dancers were not given specific instructions about the movements they should perform, they did change their decision-making behavior. In particular, it was noted that specific CI action patterns emerge as affordances of the distance between the two pelvises. We also observed significant differences between the configurations performed in the three different conditions. With IC2, dancers performed more configurations related with walking, while other skills not specific to CI, such as jumps, attracted the system of the pair of dancers. A specific type of lift, involving highly stable positions of the dancer/base (on all fours or lying positions) and with the dancer/flyer touching the floor, also emerges. The spontaneous emergence of dancer–dancer coordinative states suggests that practice should rely less heavily on explicit verbal instructions from the teacher or on the copying of models.

The most salient movements that emerged under instructional task constraints obviously form a nucleus around which more diverse and brief movement forms emerge. Hence, it would be of key importance to identify which ICs are most likely to induce and increase the frequency of occurrence of these movements that are salient for CI. This is especially relevant in relation to novice learners, who having learned these fundamental movements would then be able to develop them into more diverse and specific whole body configurations.

As Hristovski, Davids, Araujo, and Passos (2011) propose, teaching-learning processes could be directed toward creating a variety of learning situations in which dancers would themselves explore, discover, and, therefore, adapt to the information-movement demands of interpersonal coordination. These learning situations would consist in manipulating the constraints on the system's dynamics. In this article, we have considered the possibility of changing the distances between specific parts of the dancers' bodies, although the system could be constrained in an infinite number of ways depending on the aim of the task. For

example, some teachers invite dancers to explore the feeling of dancing by manipulating the fascias of the partner, instead of their muscles or bones. This instruction helps dancers to lead their partners in such a way that a spiraling movement is produced, this being a technique closely associated with CI.

Some scholars have indicated that the partner influences dance in CI (Torrents et al., 2010), and also that very strong constraints limit the emergence of varied patterns (Torrents, Castañer, & Anguera, 2011). Another kind of constraint not related with the description of the task but that is widely used in dance involves metaphorical instruction. Researchers have provided anecdotal and statistical evidence of how images support various dance movements, skills, or the creative behavior of dancers (Castañer et al., 2009; Heiland & Rovetti, 2013; Nordin & Cumming, 2006; Nordin & Cumming, 2007; Overby, 2011; Sacha & Russ, 2006; Sawada, Mori, & Ishii, 2002; Torrents et al., 2013). All these strategies are more closely related with learner-centered styles, so we agree with the view that to be effective, teachers should master multiple teaching styles, especially those related to active engagement by students (Kulinna & Cothran, 2003).

In all three conditions, the most commonly used movements when dancing CI were leading the partner and lifts. In a recent study of expert CI dancers Torrents et al. (2010) similarly found that leading the partner and lifts were the most widely used movements along with walking and turns. In the present study, dancers walked and turned especially when dancing WIC and under IC2.

In terms of the emergence of creative behavior the dancers produced a greater long-term exploratory breadth ($<q_{stat}>$) when dancing under IC2. This instruction forced the dancers to explore very different ways of dancing, because the free practice of CI is usually focused on making contact with the partner as much as possible, and especially with the center of mass of the body. This result is in agreement with the proposal of Stokes (2008, 2009), promoting search in different parts of the problem space and risk responses by using constraints. This probably accounts for the finding that although dancing WIC produced a greater exploratory breadth than did dancing under IC1, the differences between the two conditions were not significant, even though the effect size was high. However, it is interesting to note that a constraint focused on increasing the contact points (IC1) produced less creative behavior. The results from the cross-correlation analysis would also point in this direction, since they showed the effect that the partner has on the short-term coupling properties (lag 0 and 1) of the duet. The possible effects that the partner may have on the exploratory breadth of movement is an interesting topic for future research.

Given the consistency of our experimental results we further hypothesize that this model (Hristovski et al., 2011, 2013) could be useful for studying the emergence of creative behavior in any sport activity, and also for exploring the influence of ICs on the teaching-learning process of any technical movement in dancing or in sport. For example, we could study the emergence of technical-tactical behavior in soccer or in collective sports modifying task constraints during the game. In soccer the most common way of modifying task constraints is to use small-sided games, a kind of training drill that could be recorded and analyzed using the methodology used in this study. Further research is obviously needed to clarify whether the results obtained here are reproducible in other

fields related with physical exercise, and also to explore whether similar results would be obtained with persons not familiarized with CI. First results about this topic show that dancing with pelvis far helps the practitioners not familiarized with CI to explore more, but these ICs are not enough to produce the emergence of specific CI movement forms in novice dancers (Torrents, Ric, & Hristovski, in press).

A number of limitations of this study should be considered. First, many discourses on creativity have not been discussed in this article (see Banaji, Burn, & Buckingham, 2006). Second, the analysis is focused on observable motor behavior, whereas it would also be necessary to consider the influence of other behavioral factors such as participants' emotions, thoughts, or the subjective meaning of their behavior. Finally, the small sample size limits the generalizability of the results, although it should be acknowledged that this was not a key objective of the study; rather, our aim was to explore the possibility of using this model to study creative behavior or the effect of instructions regarding any other physical practice.

Conclusions

The first conclusion to be drawn from the results is that creative behavior can be encouraged by manipulating instructional task constraints. ICs that produce a very different behavior to that most commonly used by performers can lead to greater exploration of movement possibilities. By contrast, constraints that reinforce the existing tendency of performers may limit the amount of exploratory behavior. Our findings suggest that decision-making behavior is best considered at the level of the performer-environment relationship, such that it is viewed as emerging over time from the interactions of individuals with environmental constraints as they seek to achieve specific functional goals.

The results of this study also show that it is possible to stimulate the practice of specific technical skills simply by modifying ICs, without those skills having to be isolated from the normal practice of the activity. An approach of this kind would have potential in other fields of physical education and sport, where rather than memorizing a sequence of actions players need to develop their capacity to detect informational constraints that are specific to goal-paths (Davids, Button, & Bennett, 2008).

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